

2(a)	any two from: 1 data, are ordinal / can be ranked ; 2 data points within samples are independent of each other ; 3 there are 20 paired observations ; 4 coffee farms / sampling points, were selected at random ; 5 AVP ;	2										
2(b)	(there is) no correlation between the water availability and the number of months that elephants entered each (coffee) farm per year ;	1										
2(c)(i)	all correct ; <table><tr><th>r_s value</th><th>significance</th></tr><tr><td>0.63</td><td>yes</td></tr><tr><td>-0.30</td><td>no</td></tr><tr><td>0.02</td><td>no</td></tr><tr><td>0.38</td><td>no</td></tr></table>	r_s value	significance	0.63	yes	-0.30	no	0.02	no	0.38	no	1
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0.63	yes											
-0.30	no											
0.02	no											
0.38	no											

Question	Answer	Marks
2(c)(ii)	any one from: negative correlation / the greater the density of December trees, the fewer the number of months that elephants entered each (coffee) farm / ora and (correlation) is weak / is not significant ; there is no correlation (between the density of December trees and the number of months that elephants entered each farm) ;	1
2(d)	any three from: <i>support:</i> <i>idea that</i> reducing water availability is (significantly) correlated with a reduced number of times that elephants enter the (coffee) farm ; <i>idea that</i> water availability was the only variable found to be significant (so reducing the number of water bodies might be the most effective action) ; ora <i>does not support (max 2):</i> - water bodies only counted, once / in March / in 2008 ; - investigation only, in one geographical area (of India) / at Nagarahole National Park ; <i>idea that</i> size of, ponds / lakes / water holes, might also be important ; <i>idea that</i> (questionnaire) does not record actual, number of times elephants entered the farms / number of elephants (per month) ; <i>idea that</i> correlation does not indicate causation (so reducing the availability of water may not affect the number of elephant visits) ;	3